MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6BQ5

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Miniature, Beam Power, pentode
Outline --- 6-4 (EIA)
Base --- E9-1
Envelope --- T6-1/2
Cathode --- Coated unipotential
Base connections:

Pin No.:	1	2	3	4	5	6	7	8	9
Element:	ic	g1	k, g3	h	h	ic	a	ic	g2

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ebb	Ec1	Ec2	Ecc2	Rg1	Ik	Pp	Pg2	pg2	Ehk	Alt.
Unit:	V	Vdc	Vdc	Vdc	Vdc	Vdc	Meg	mAdc	W	W	w	V	Ft.
Maximum:	6.9	300	550	0	300	550	1.0	65	12	2.0	4.0	±100	10,000
Minimum:	5.7	---	---	-100	---	---	Note 1	---	--	---	---	---	---
TEST CONDITIONS:	6.3	250	---	-7.3	250	---	---	---	--	---	---	---	---

GENERAL:

Qualification - Not required.

Method	Requirement or test	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits			Unit
						Min	Bogey	Max	
	Quality conformance inspection, part 1								
1266	Total grid current		0.65	II	Ic1	0	---	-2.0	uAdc
1256	Electrode current (1) (anode)	Eb=60 Vdc; Ec1=0 Ec2=250 Vdc; Rg1=2.2 Meg; (see note 2)	0.65	II	Ib	53	---	---	mAdc
1201	Shorts and discontinuity detection		0.4	II	---	---	---	---	---

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Method	Requirement or test	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits			Unit
						Min	Bogey	Max	
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	---	4.0	S3	R	100	---	---	Meg
1301	Heater current		4.0	S3	If	685	760	835	mA
1336	Heater-cathode leakage	---	4.0	II	Ihk:	---	---	50	uAdc
1256	Electrode current (2) (anode)	Ef=5.7V; Eb=60Vdc; Ec1=0; Ec2=250Vdc; Rg1=2.2 Megs; (see notes 2 and 3)	4.0	S3	Ib	---	---	10	%
1306	Transconductance		4.0	S3	Sm	9100	11300	13500	umhos
1341	Power output	Esig=4.7 Vac; Rp=5200	4.0	II	Po	5.2	6.0	---	W
1246	Audio frequency noise	Esig=550 mVac; Ebb=250 Vdc; Ecc2=200 Vdc; Rg1=0.1 Meg; Rg2=0.022 Meg; Cg2=4 uf; Rp=5000 Ω; Ec1= -6 Vdc; (see note 4)	2.5	II	---	---	---	---	
1256	Electrode Current (3) (anode)		4.0	S3	Ib	35	48	65	mAdc
1256	Electrode Current (Screen)		4.0	S3	Ic2	---	5.5	11.0	mAdc
---	Current ratio:	Eb=60 Vdc; Ec1=0; Ec2=250 Vdc; Rg1 = 2.2 Meg; (see note 2)	4.0	II	Ib/Ic2	4:1	---	---	---
1121	Base strain		---	---	---	---	---	---	---
2126	Glass strain		2.5	I	---	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---	---

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Method	Requirement or test	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits			Unit
						Min	Bogey	Max	
	Quality Conformance inspection, part 3								
---	Intermittent life test:	Group A; Ebb=300 Vdc; Ecc=300 Vdc; Rk=150 Ω; Rg=2.2 Meg; Rp=1000 Ω; Ehk= +200 Vdc	---	---	t	500	---	---	Hrs
---	Intermittent life test End points:	Power output	---	---	Po	3.0	---	---	W
		Grid #1 current	---	---	Ic1	0	---	-6.0	uAdc

NOTES:

1. The 1.0 megohm rating applies for self bias operation. For fixed bias operation the maximum allowable value is 0.3 megohm.
2. Apply Eg1 to the grid through a series capacitor of 0.01 uf. Eg1 is a 90 V peak to peak square wave at approximately 6 kHz such that the tube under test is conducting for exactly one-half time and is cutoff for one half the time. Ib and Ic2 as read on DC meters shall be within the limits specified herein.
3. The change of electrode current (anode) resulting from the change in filament voltage shall not exceed the limit specified herein.
4. The rejection level shall be set at the VU meter reading obtained during calibration.

User activities:

Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

(Project 5960-N-876)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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